Extremes for metocean: Covariates

Philip Jonathan

Lancaster University

IOGP, London, September 2026
(slides at *www.lancs.ac.uk/∼jonathan)*



Overview of masterclass

- A: Fundamentals
- B: Covariates
- C: Multivariate
- D: covXtreme
- E: Recent developments & take-aways

Overview of B: Covariates

- Motivating thought experiment
- Motivating application
- Overview of analysis procedure
  - Frequentist: maximum penalised likelihood, cross-validation, bootstrapping
  - Bayesian: priors, Markov Chain Monte Carlo
- Typical results
- Key issues

Sanity check

All models are wrong, some models are useful

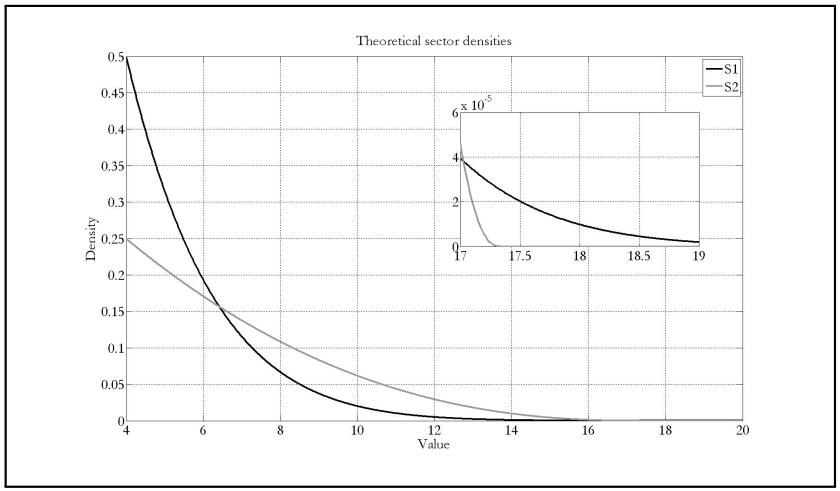
Consistent balance of engineering and statistical insights

A motivating "toy" thought experiment

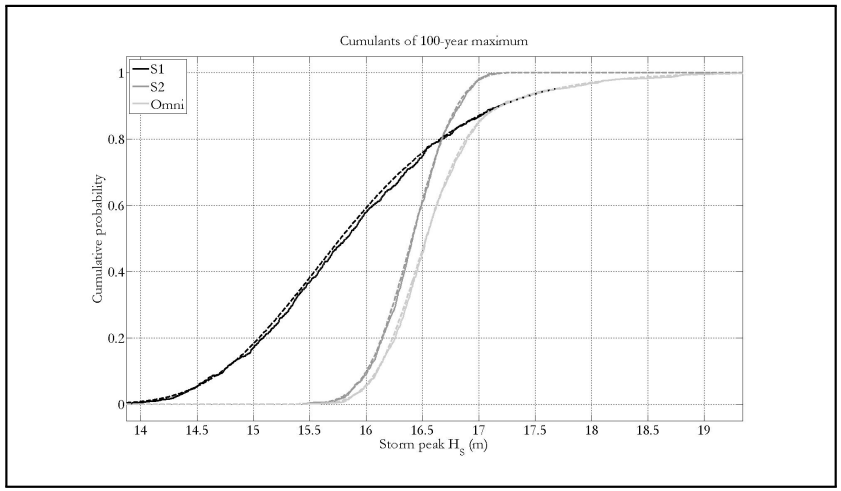
Question: Do we get return values wrong if we ignore directionality of extremes?

- A directional extremes problem
- Two homogeneous 180° directional sectors S_1 and S_2, extremes are GP-distributed in each
- ξ, σ and u values all potentially different between sectors
- Random samples of size 1250 from each sector corresponding to 25 years
- Test
  - "Null hypothesis" H_0: $\xi_1 = \xi_2$, $\sigma_1 = \sigma_2$
  - "Alternative hypothesis" H_A: $\xi_1 \neq \xi_2$, $\sigma_1 \neq \sigma_2$
- We will consider case (ξ_1,σ_1,u_1)=(-0.1,2,4) and (ξ_2,σ_2,u_2)=(-0.3,4,4)
  - See Jonathan et al. [2008] for more

Tail characteristics



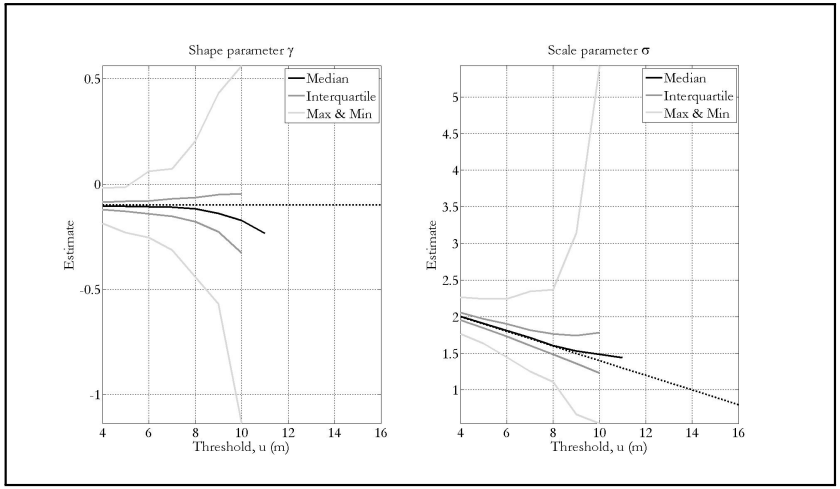
Sector densities (theory)



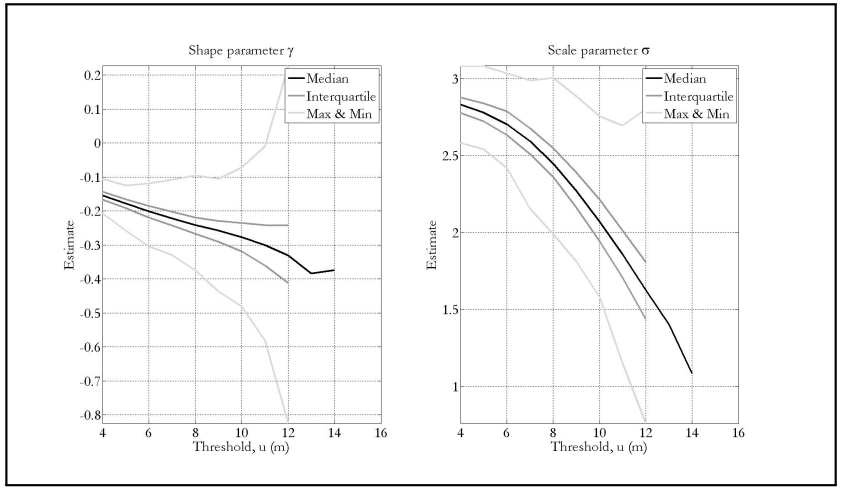
100-year maximum

- S1 (black) dominates far tail, but S2 (grey) prevalent for values $\in [7, 14]$
- Typical sample will have more examples from S2 than S1, causing issues
- 100-year maximum dominated by S2 below 17, and by S1 above

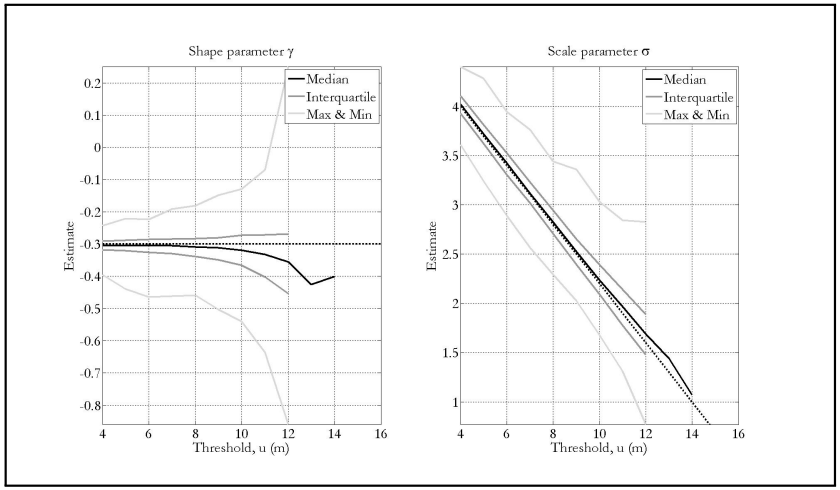
Parameter estimates with threshold



H_A: S_1 estimates with threshold



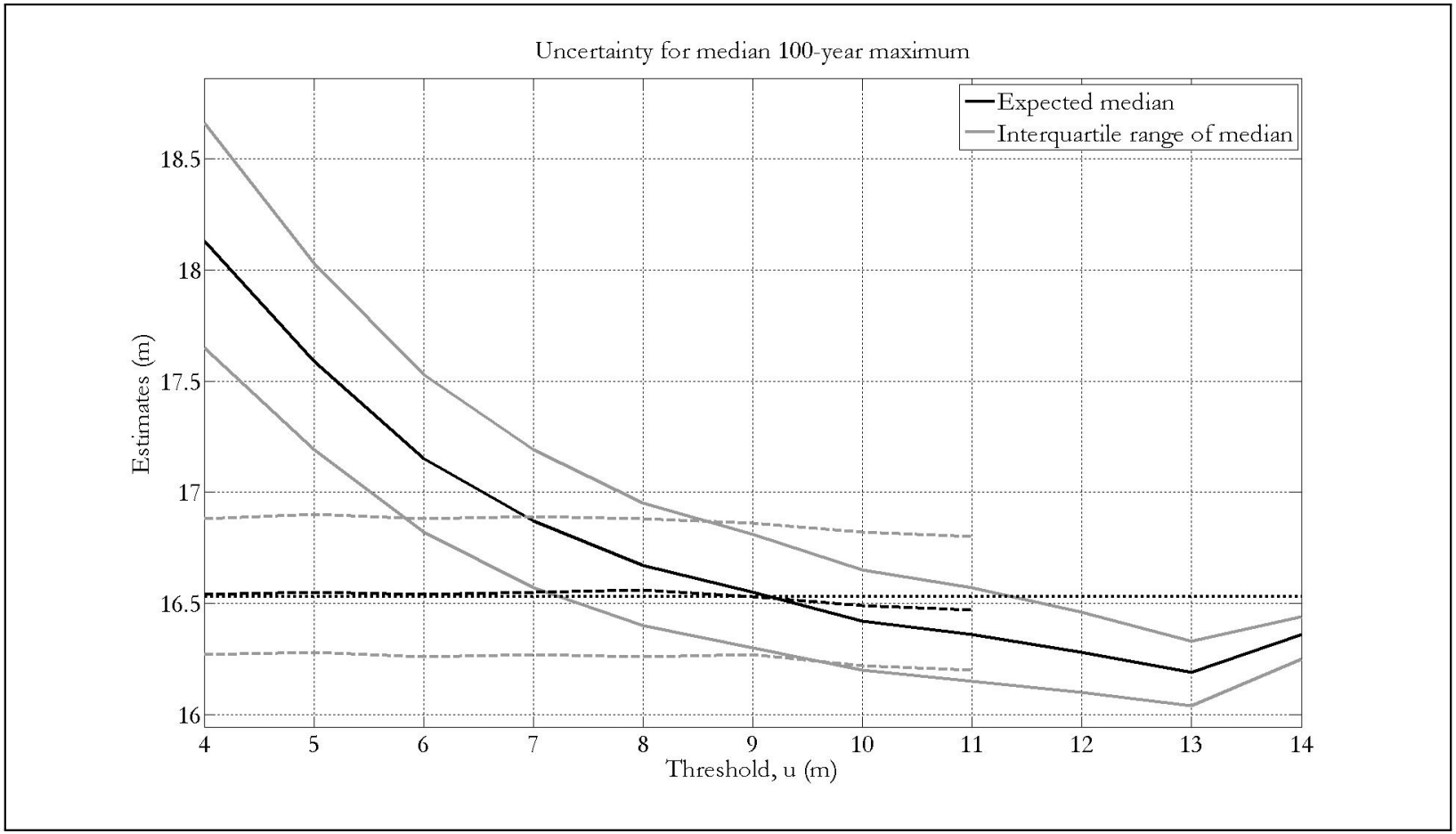
H_0: estimates with threshold



H_A: S_2 estimates with threshold

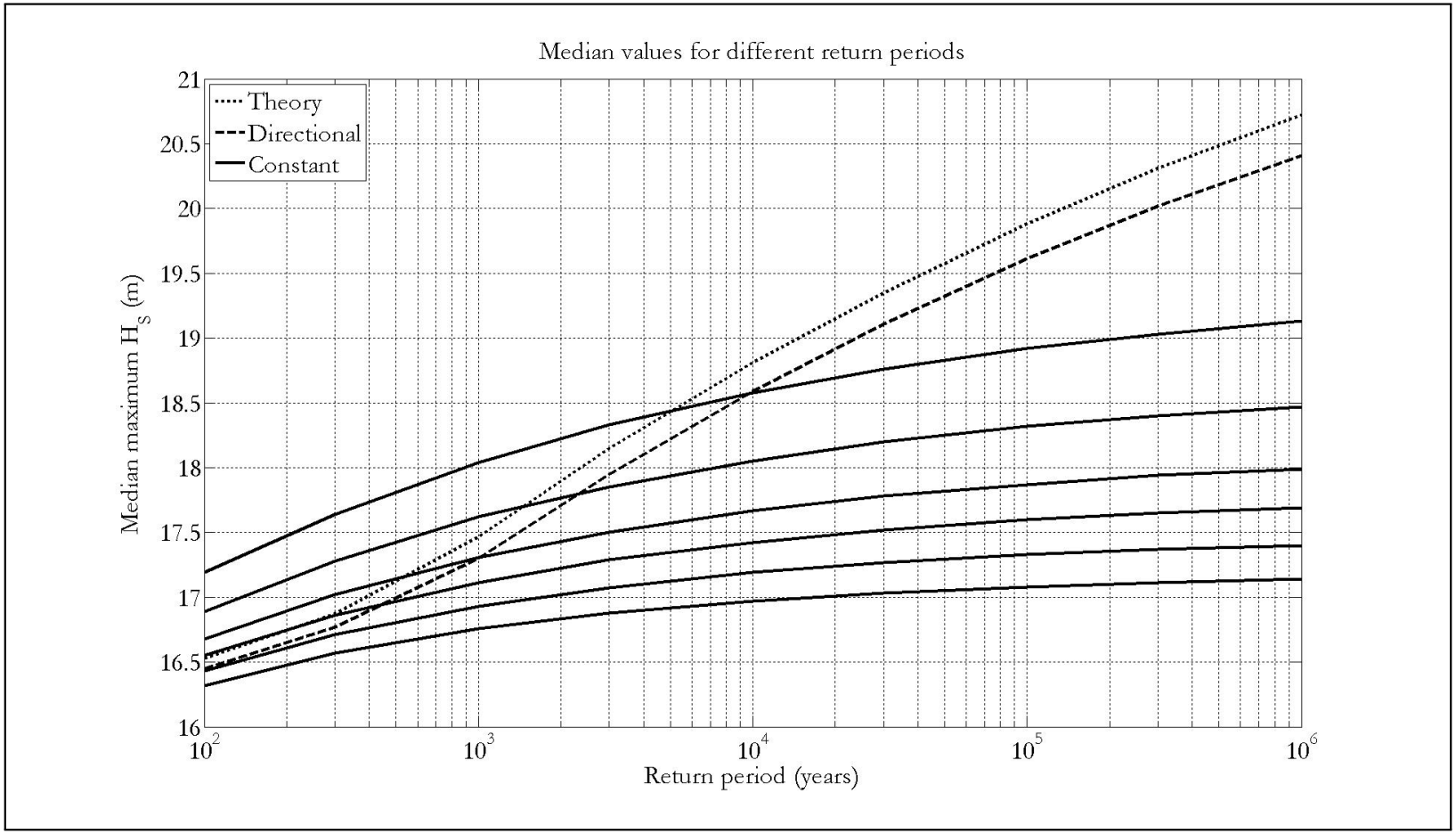
- H_A looks OK in S_1 and S_2
  - $\hat{\xi}$ constant with u, $\hat{\sigma}$ linear
- H_0 suspicious!
  - $\hat{\xi}$ NOT constant with u

Estimated median 100-year maximum



Median 100-year event: true (dot), H_A (dash), H_0 (solid)

Implications for long return periods



Median P-year event: true (dot), H_A (dash),

H_0 (solid, for thresholds 6,7,...,11m)

In other words ...

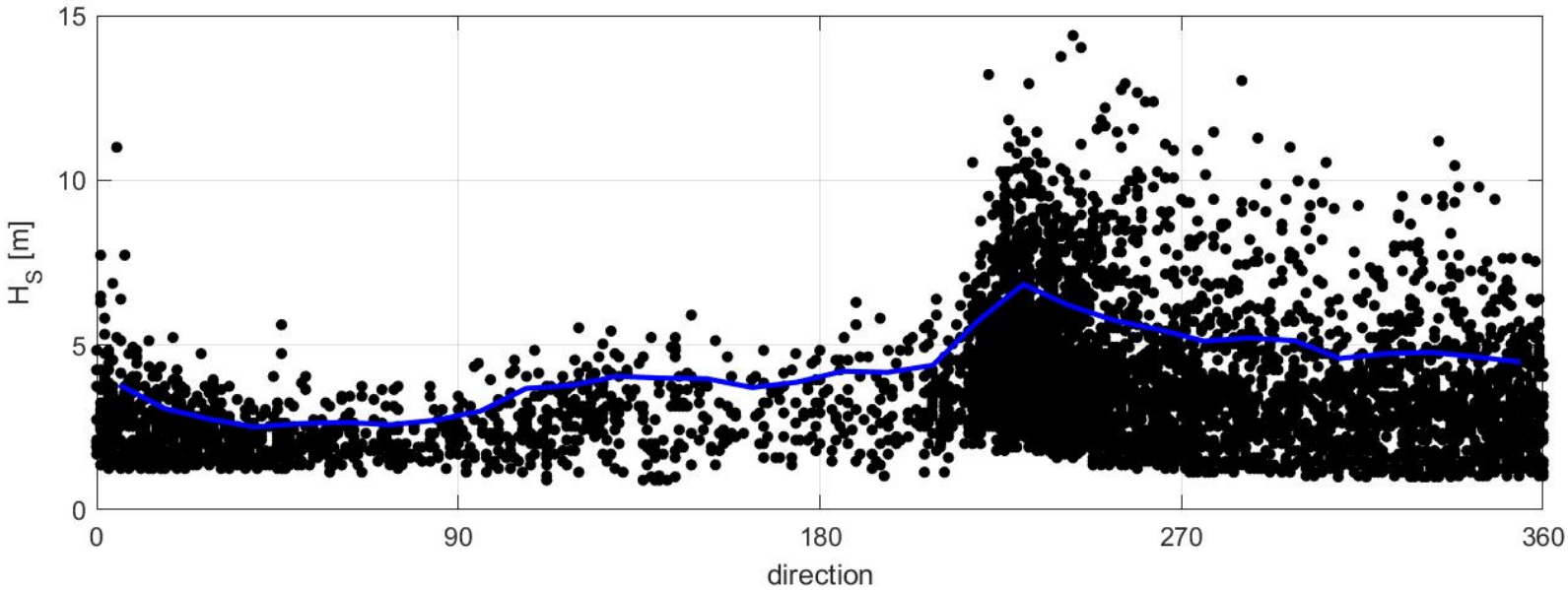
If we misspecify the extreme value model, results are unpredictable. We honestly don't know whether return value estimates will be biased high, low or be unbiased. Threshold choice becomes a bigger problem. Ignore covariates at your peril in general

If we are confident that the far tail is dominated by one direction, it makes sense to look at that direction in isolation to estimate the omni-directional return value, provided we have enough data from that direction

If we want consistent estimates of return values across every directional sector, we must fit an appropriately-specified directional extreme value model. We need directional-seasonal covariate models for quantities like H_S; and similar logic to include other important covariates

Motivating application

- Typical data for northern North Sea
  - Storm peak H_S ($\triangleq H_S^{sp}$) on direction
  - Rate and size of occurrence varies with direction



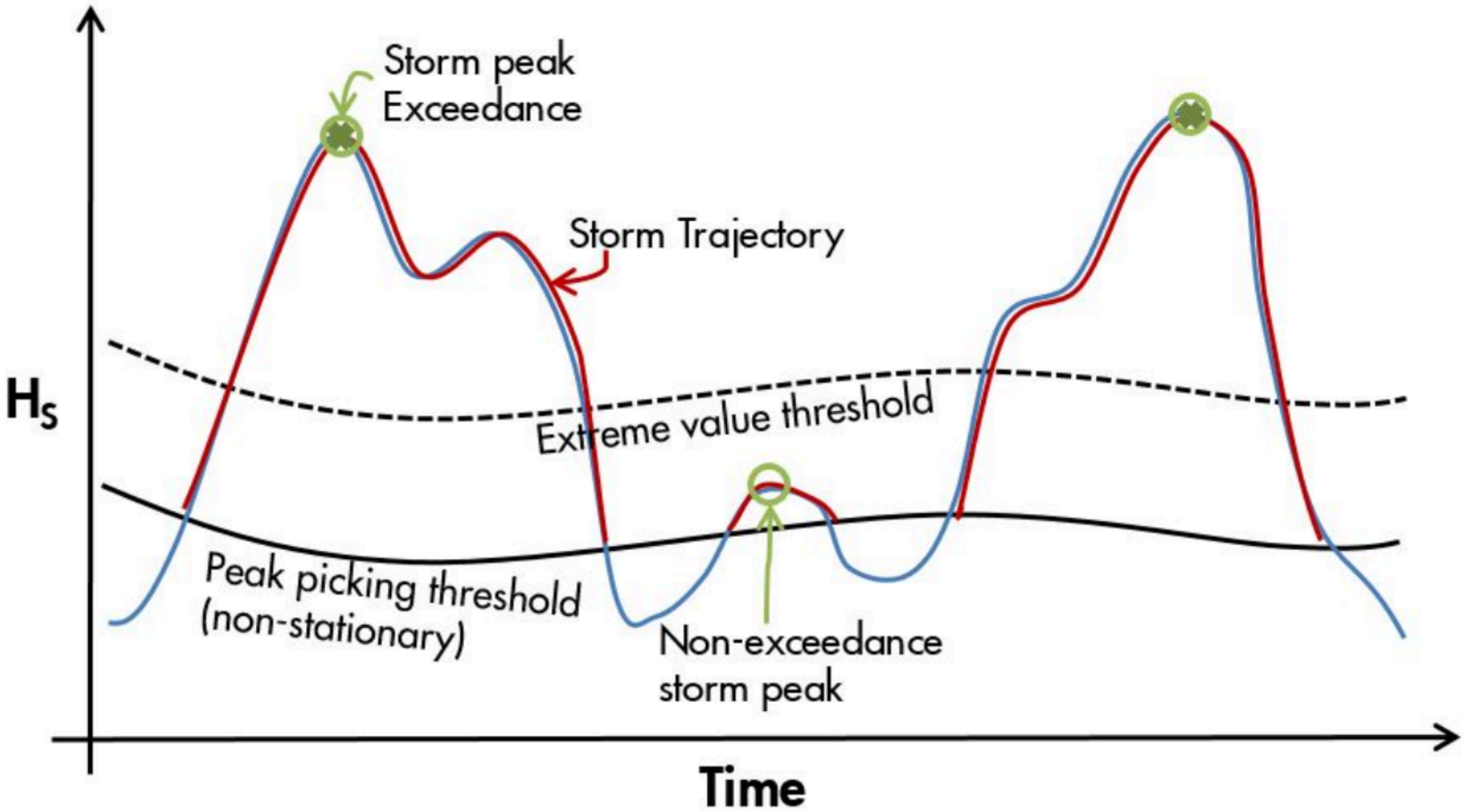
Storm peak H_S^{sp} on direction, with $\tau = 0.8$ extreme value threshold

- Environmental extremes vary smoothly with multidimensional covariates
  - We will use only direction for simplicity here

Full analysis procedure

- Input
  - Time-series of sea state H_S and associated variables (e.g. T_P, surge, wind)
- Preprocessing ("peak picking")
  - Isolate storms
  - Isolate storm peak quantities (H_S^{sp} and associated)
  - Isolate accompanying multivariate storm trajectories
- Key statistical models (all directional)
  - Model for distributional quantiles, used as extreme value thresholds u
  - Model for rate of occurrence of exceedances of threshold u per annum
  - Model for size of occurrence of exceedance of threshold u
- Metocean inferences
  - Use fitted models to estimate return values for H_S^{sp} and associated storm peak quantities
  - Combine simulated storm peak quantities with storm trajectories to estimate return values for sea-state quantities and individual waves

Key quantities



Key variables describing a storm. Peak-picking threshold (solid black), extreme value threshold (dashed black), storm peak (green circle), storm peak over threshold (green star), storm trajectory (red)

Managing computational complexity of parameter variation with direction

- Periodic domain $\mathcal{D}$ ($= [0, 360)^\circ$ say) for direction θ
- Partition directional domain $\mathcal{D}$ into m bins of equal width $d > 0$ and bin centres $\boldsymbol{\theta} = \{\theta_j\}_{j=1}^m$
- Index set $\mathcal{I}$ maps each storm peak observation i onto one of the bins j; so $\theta_{\mathcal{I}(i)}$ is the bin centre direction for observation i
- Any model parameters $\eta(\theta)$ for $\theta \in \mathcal{D}$ are also evaluated at bin centres only, and can be written as a vector $\boldsymbol{\eta} = \{\eta_j\}_{j=1}^m = \{\eta(\theta_j)\}_{j=1}^m$
- To keep things simple, we might set $m = 360$ so that each direction is specified to the nearest degree
  - But this might not always be the wisest choice, for physical and especially computational reasons

A model for extreme value threshold, $u(\theta)$

- Gamma model for **full distribution** of storm peaks X with $\theta \in \mathcal{D}$
- Density for a single storm peak observation x

$$f_{\text{Gmm}}(x|\alpha,\kappa) = \frac{\kappa^{\alpha}}{\Gamma(\alpha)} x^{\alpha-1} \exp(-\kappa x)$$

  with parameters shape $\alpha > 0$ and scale (or rate) $\kappa > 0$
- Sample likelihood for storm peak values $\boldsymbol{x} = \{x_i\}_{i=1}^{n}$ with associated directional bins $\{\theta_{\mathcal{I}(i)}\}_{i=1}^{n}$ is

$$\mathcal{L}_{\text{Gmm}}(\boldsymbol{\alpha},\boldsymbol{\kappa}|\boldsymbol{x}) = \prod_{i=1}^{n} f_{\text{Gmm}}(x_i; \alpha_{\mathcal{I}(i)}, \kappa_{\mathcal{I}(i)})$$

  for parameter vectors $\boldsymbol{\alpha} = \{\alpha_j\}_{j=1}^{m}$ and $\boldsymbol{\kappa} = \{\kappa_j\}_{j=1}^{m}$
- α and κ vary smoothly with direction on $\mathcal{D}$

- Quantiles of the gamma distribution define extreme value thresholds u. We consider model averages over multiple thresholds
- Obvious alternative approach is quantile regression

A model for expected annual rate $\rho(\theta)$ of threshold exceedance

- Poisson model for annual rate $\rho(\theta)$ of threshold exceedance for $\theta \in \mathcal{D}$
- Count the number C of storm peak threshold exceedance counts $\boldsymbol{c} = \{c_j\}_{j=1}^{m}$ in the directional bins
- Density for a single directional bin j

$$f_{\text{Pss}}(c_j|\rho_j) = \frac{(d\rho_j)^{c_j}}{c_j!} \exp\{-d\rho_j\}$$

  with rate $\rho \geq 0$, where $d\rho_j$ is the expected number of threshold exceedances in the bin
- Sample likelihood

$$\mathcal{L}_{\text{Pss}}(\boldsymbol{\rho}|\boldsymbol{c}) = \prod_{j=1}^{m} f_{\text{Pss}}(c_j|\rho_j)$$

  for parameter vector $\boldsymbol{\rho} = \{\rho_j\}_{j=1}^{m}$
- ρ varies smoothly with direction on $\mathcal{D}$

A model for the size X' of threshold exceedances

- GP model for storm peaks X' over threshold $u(\theta)$ with direction $\theta \in \mathcal{D}$
- Density for a single storm peak threshold exceedance x'

$$f_{\text{GP}}(x'|\xi, \nu, u) = \frac{1}{\sigma}\left(1 + \frac{\xi}{\sigma}(x' - u)\right)_+^{-1/\xi - 1} \quad \text{with } \nu = \sigma(1+\xi)$$

  for $x' > u$ with parameters shape $\xi > -1$ and adjusted scale $\nu > 0$
- Sample likelihood for storm peak threshold exceedances $\boldsymbol{x}' = \{x'_{i'}\}_{i'=1}^{n'}$ with associated directional bins $\{\theta_{\mathcal{I}(i')}\}_{i'=1}^{n'}$ is

$$\mathcal{L}_{\text{GP}}(\boldsymbol{\xi}, \boldsymbol{\nu}|\boldsymbol{x}', \boldsymbol{u}) = \prod_{i'=1}^{n'} f_{\text{GP}}(\boldsymbol{x}'; \xi_{\mathcal{I}(i')}, \nu_{\mathcal{I}(i')}, , u_{\mathcal{I}(i')})$$

- ξ and ν (and u) vary smoothly with direction on $\mathcal{D}$

In other words ...

We do extreme value analysis of storm peaks (assumed independent given covariates). These are isolated from the time-series using "peak-picking"

Then we estimate a model for the extreme value threshold $u(\theta)$. Here we use a gamma distribution, with parameters varying with direction, estimated by maximum likelihood estimation

Then we estimate a model for the RATE of threshold exceedance $\rho(\theta)$. Here we use a Poisson distribution, with parameters varying with direction, estimated by maximum likelihood estimation

Then we estimate a model for the SIZE of threshold exceedance. This is a GP, with parameters varying with direction, estimated by maximum likelihood estimation

The three models combined give us a way of simulating storm peak events corresponding to very long return periods

Next we discuss HOW to represent the smoothly-varying parameters in these models using splines

Smooth covariate representations in 1-D

Covariate representation

- Index set $\mathcal{I}$ of m directional bins on $\mathcal{D}$; each observation belongs to exactly one bin
- On $\mathcal{I}$, assume model parameter $\eta_j = \sum_{k=1}^{p} B_{jk}\beta_k$, $j = 1, 2, ..., m$, or

$$\boldsymbol{\eta} = \boldsymbol{B}\boldsymbol{\beta}, \quad \eta \in (\xi, \nu, \rho, u)$$

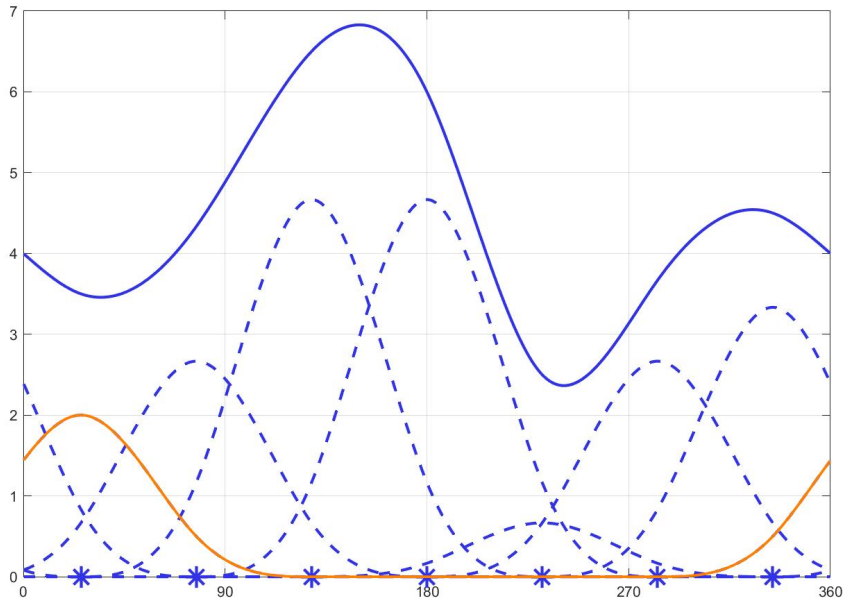
- $\boldsymbol{B} = \{B_{jk}\}_{j=1;k=1}^{m;p}$ set of p basis vectors for $\mathcal{D}$ (pre-specified usually)
- $\boldsymbol{\beta} = \{\beta_k\}_{k=1}^{p}$ basis coefficients (to be estimated)
- Inference reduces to estimating $\boldsymbol{\beta}$ (and maybe $\boldsymbol{B}$) for each η

Roughness penalty

- $\boldsymbol{D}$ is a $p \times p$ (wrapped) differencing matrix
- $\boldsymbol{D}\boldsymbol{\beta}$ is a vector of differences between consecutive values of $\boldsymbol{\beta}$
- $\boldsymbol{\beta}'\boldsymbol{D}'\boldsymbol{D}\boldsymbol{\beta}$ is the scalar sum of squares difference, a roughness penalty!

P-splines

- p regularly-spaced knots on $\mathcal{D}$
- $\boldsymbol{B}$ consists of p B-spline bases
  - Order d
  - Each using $d+1$ consecutive knot locations
  - **Local support**
  - Wrapped on $\mathcal{D}$
  - Cox - de Boor recursion formula



Periodic P-splines

- p is fixed and "over-specified"
- Knot locations fixed
- Roughness $\boldsymbol{\beta}'\boldsymbol{D}'\boldsymbol{D}\boldsymbol{\beta}$ penalised

- Alternatives: Fourier, Voronoi partition, adaptive splines, ...

Maximised penalised likelihood

- Choose $\boldsymbol{\eta}$ to maximise

$$\mathcal{L}^*(\boldsymbol{\eta}) = \mathcal{L}(\boldsymbol{\eta}) \exp\left(-\lambda \boldsymbol{\beta}' \boldsymbol{D}' \boldsymbol{D} \boldsymbol{\beta}\right)$$

  for given roughness coefficient $\lambda > 0$
- Use cross-validation to estimate the value of λ which maximises predictive performance
- Use bootstrapping to estimate uncertainties on everything of interest

- General procedure: works for all of u, ρ, ξ and ν
- Using a simple piecewise constant covariate representation, this is exactly the idea behind covXtreme

- This is a "frequentist" approach; alternative is Bayesian inference

Bayesian inference

Priors

$$\begin{aligned} \text{prior density of } \boldsymbol{\beta} &\propto \exp\left(-\lambda \boldsymbol{\beta}' \boldsymbol{D}' \boldsymbol{D} \boldsymbol{\beta}\right) \\ \text{prior density of } \lambda &\sim \text{gamma} \end{aligned}$$

Inference for GP

- Objective: sample from posterior density $f(\boldsymbol{\beta}_\xi, \lambda_\xi, \boldsymbol{\beta}_\nu, \lambda_\nu | \boldsymbol{x}')$
- Parameter set $\Omega = \{\boldsymbol{\beta}_\xi, \lambda_\xi, \boldsymbol{\beta}_\nu, \lambda_\nu\}$
- Generic conditional structure for Metropolis Hastings within Gibbs sampling from full conditionals

$$\begin{aligned} f(\boldsymbol{\beta}_\xi | \boldsymbol{x}', \Omega \setminus \boldsymbol{\beta}_\xi) &\propto f(\boldsymbol{x}' | \boldsymbol{\beta}_\xi, \Omega \setminus \boldsymbol{\beta}_\xi) \times f(\boldsymbol{\beta}_\xi | \lambda_\xi) \\ f(\boldsymbol{\beta}_\nu | \boldsymbol{x}', \Omega \setminus \boldsymbol{\beta}_\nu) &\propto f(\boldsymbol{x}' | \boldsymbol{\beta}_\nu, \Omega \setminus \boldsymbol{\beta}_\nu) \times f(\boldsymbol{\beta}_\nu | \lambda_\nu) \\ f(\lambda_\xi | \boldsymbol{x}', \Omega \setminus \lambda_\xi) &\propto f(\boldsymbol{\beta}_\xi | \lambda_\xi) \times f(\lambda_\xi) \\ f(\lambda_\nu | \boldsymbol{x}', \Omega \setminus \lambda_\nu) &\propto f(\boldsymbol{\beta}_\nu | \lambda_\nu) \times f(\lambda_\nu) \end{aligned}$$

- Inference for ρ, u similar

In other words ...

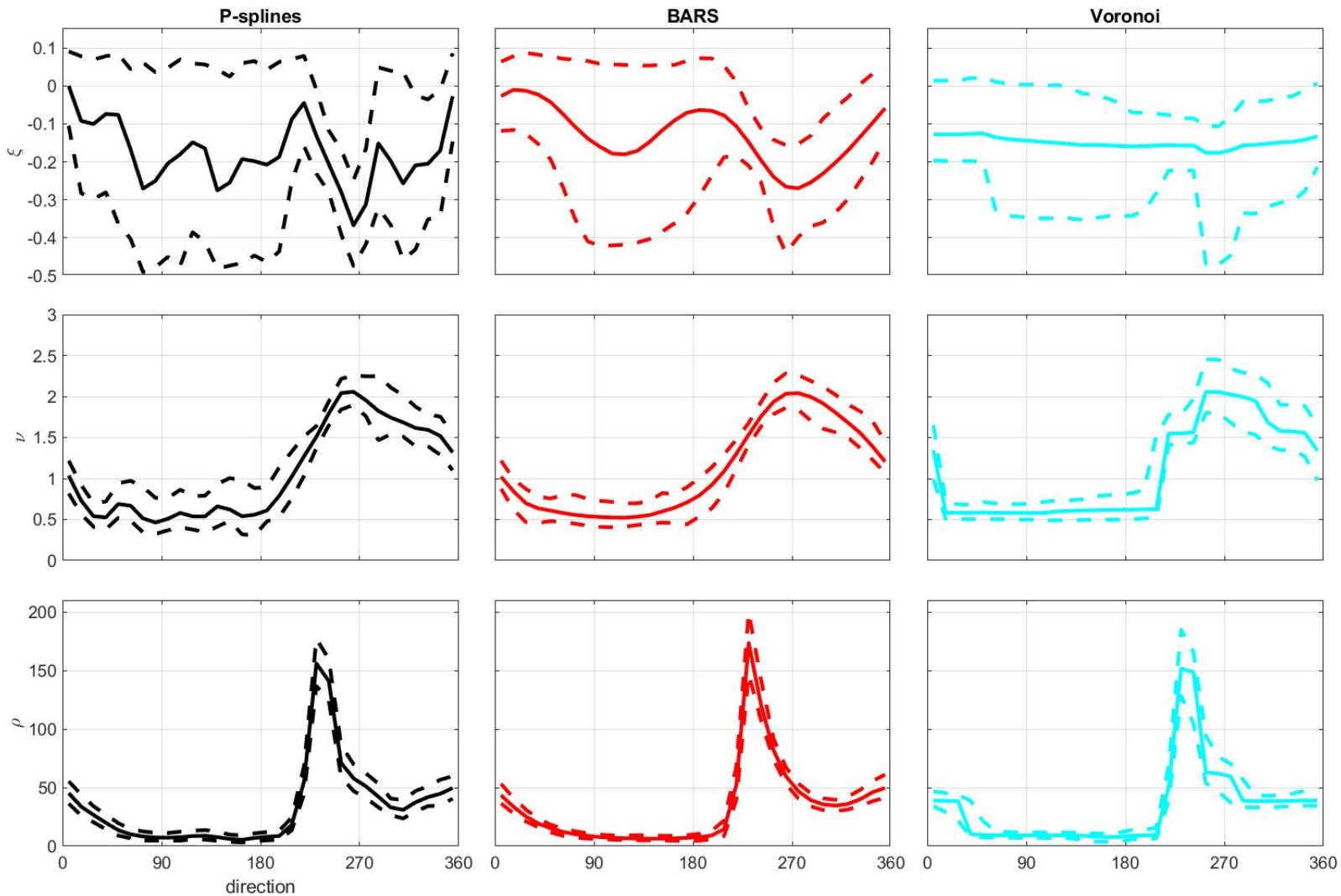
We can represent how any parameter varies on $[0, 360)$ using a periodic spline. The spline is just a linear combination of BASIS VECTORS $\boldsymbol{B}$. The weights we give to each basis vector gives the spline its shape. So we need to estimate the weights $\boldsymbol{\beta}$ (and hence the spline shape $= \boldsymbol{B}\boldsymbol{\beta}$) which best fits the data

A spline can be really wiggly and non-physical. So we put a penalty on the weights to dampen wiggliness. We use cross-validation to estimate the spline wiggliness giving BEST OUT-OF-SAMPLE predictive performance; this is an example of PENALISED maximum likelihood estimation

We also use bootstrap resampling to quantify the uncertainty in the shape of the spline. We propagate uncertainty of spline shape into subsequent calculations of return values

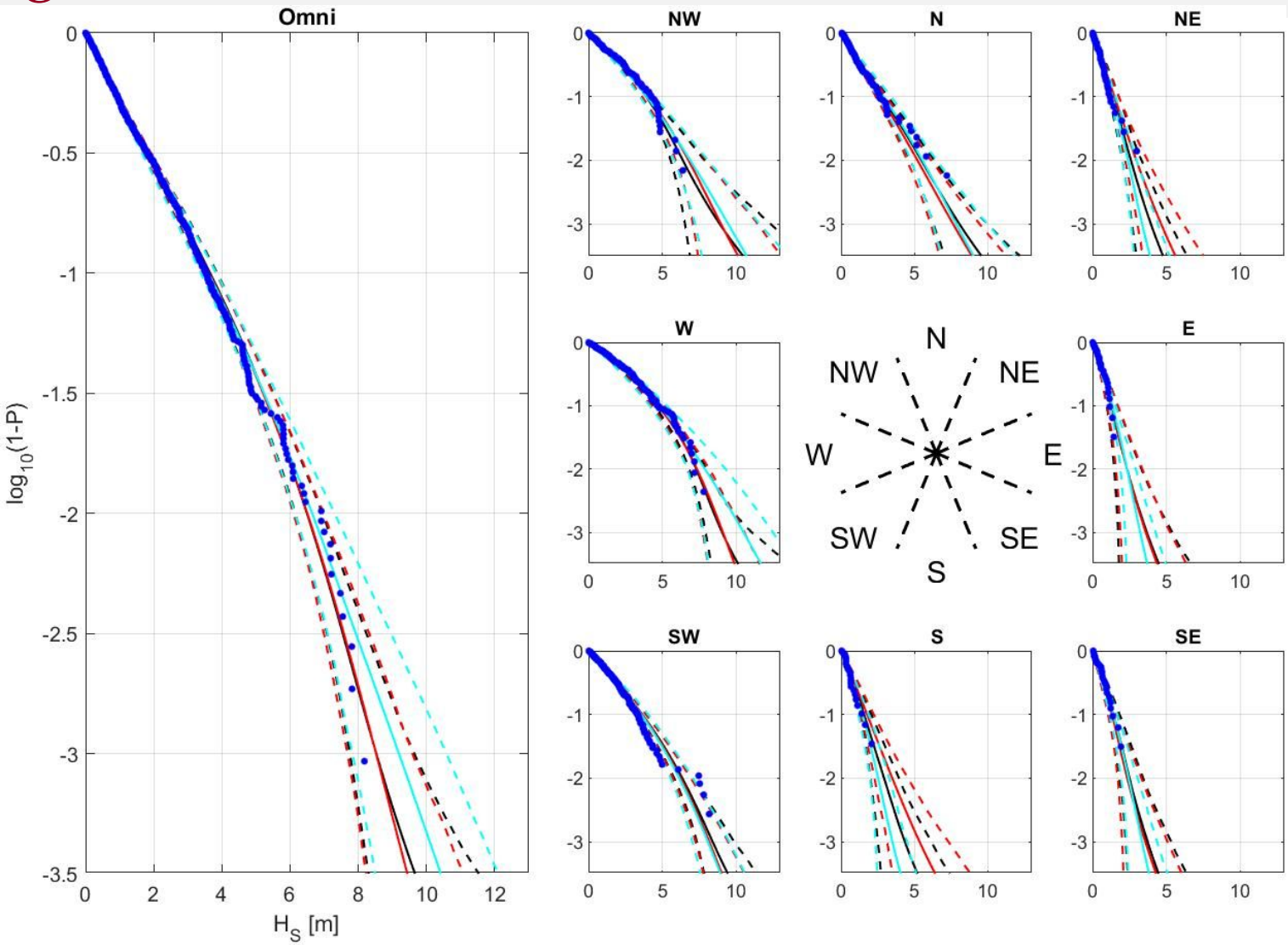
The "bees knees" way of doing this uses Bayesian inference. Popular algorithms are Metropolis Hastings and Gibbs sampling, as shown next

Posterior parameter estimates for northern North Sea



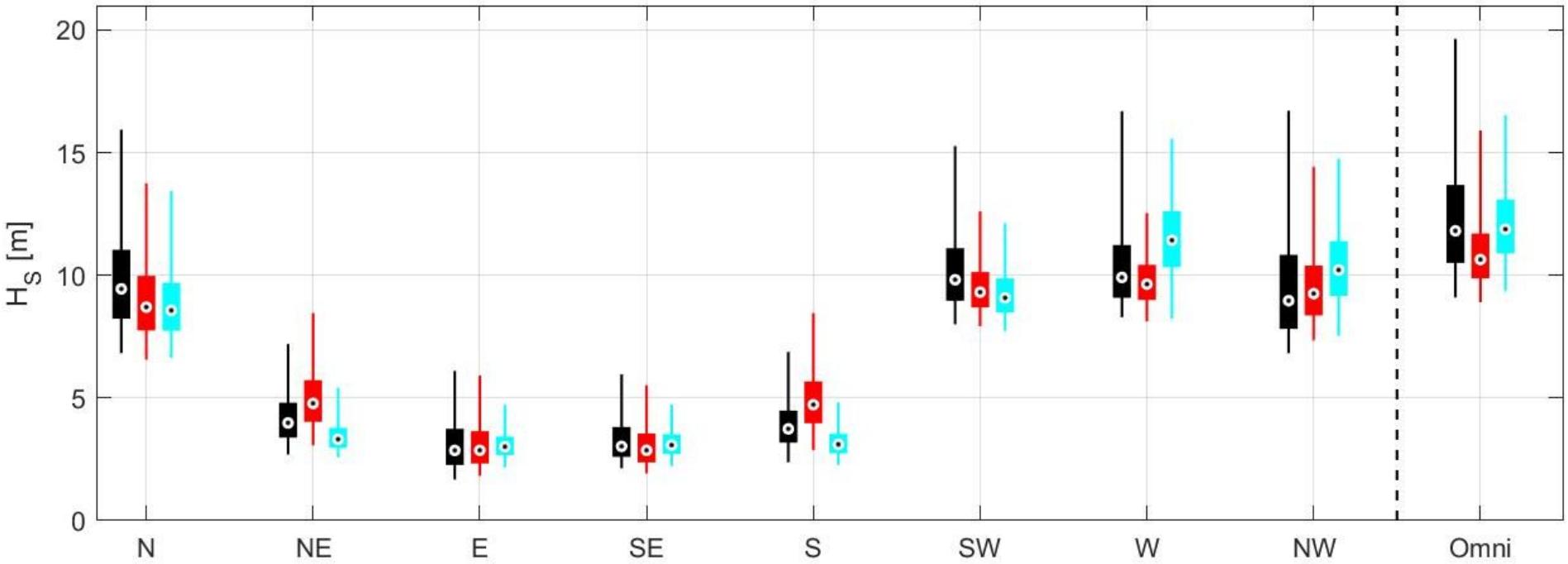
Comparing Bayesian inference using P-splines, adaptive regression splines and Voronoi partition; Zanini et al. [2020]

Fit diagnostic



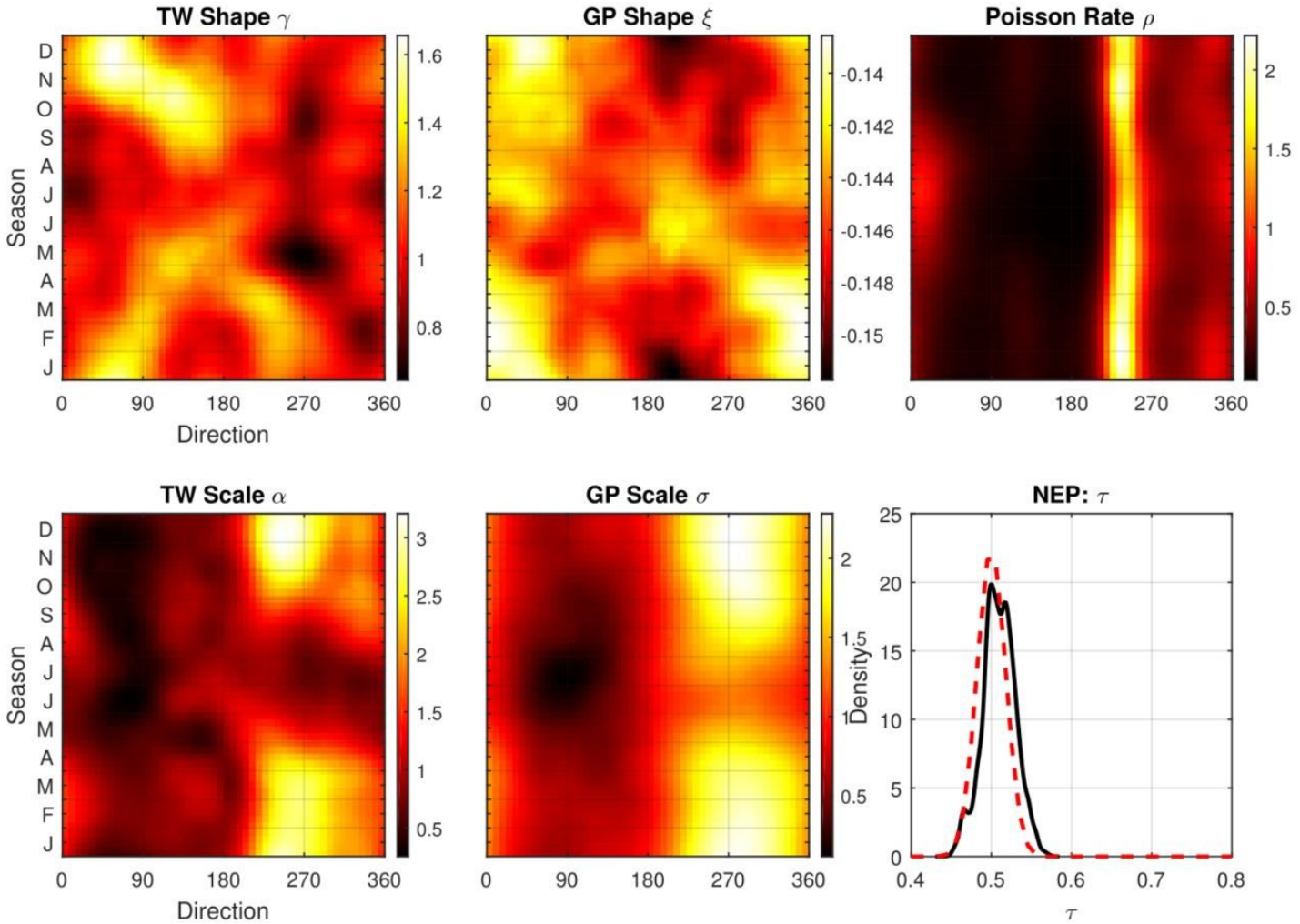
Empirical tail (blue); posterior means and 95% credible intervals for quantile levels from different models

Directional posterior predictive distribution of 1000-year maximum



- Box-whiskers with 2.5%, 25%, 50%, 75% and 97.5% percentiles

Extension to 2D : directional-seasonal



- 2-D tensor product P-spline bases for same northern North Sea location
- Marginal posterior median estimates (plus posterior density for τ)
- Uses hybrid truncated Weibull-GP model

In other words ...

We can estimate how parameters of extreme value models vary with covariates in a rigorous, logical manner. Case studies show it doesn't really matter how we do this, so long as the covariate representation is sufficiently flexible. A simple piecewise constant model does well $\Rightarrow$ covXtreme

Using MLE with cross-validation and boostrapping, or Bayesian inference, we have a way of quantifying the distribution of model parameter estimates, and propagating uncertainty through the analysis $\Rightarrow$ covXtreme

As a result, instead of "point estimates" of parameters etc., we now have DISTRIBUTIONS of PARAMETER FUNCTIONS which properly capture the non-stationarity and uncertainty in our models of the ocean environment

We need to use these distributions wisely to estimate return values! In future it would be wiser to AVOID CALCULATING RETURN VALUES at all, and move to full probabilistic design; i.e. calculating probabilities of structural failure / utility directly using our distributions, coupled to fluid loading models

Covariate modelling: what can go wrong?

- Small sample size
- Selection of appropriate non-stationary threshold(s)
  - Use diagnostics e.g. $\hat{\xi}|(u, \theta)$
  - Model averaging probably better when choice not obvious
- Complex tail structures requiring multiple covariates or very high u
- Estimation of optimal roughness coefficients λ using cross-validation in frequentist analysis can be problematic; Bayesian inference tends to behave more reasonably

Implications

- Propagate and quantify uncertainty carefully
  - No excuses!
- Model averaging in place of hyper-parameter selection
- Use cross-validation and bootstrapping in frequentist analysis; otherwise use Bayesian inference

All models are wrong, some models are useful

If you want to read more

- Key papers for spline applications: Marx and Eilers [1998], Chavez-Demoulin and Davison [2005]
- Basic ideas around covariate modelling: Jonathan et al. [2008], Jones et al. [2016], Feld et al. [2019a], Feld et al. [2019b]
- Lancaster / Shell: Randell et al. [2016] (North Sea), Ross et al. [2017] (South China Sea), Zanini et al. [2020] (North Sea)
- Lots of related work at `www.lancaster.ac.uk/~jonathan`

References

V. Chavez-Demoulin and A. C. Davison. Generalized additive modelling of sample extremes. *J. Roy. Statist. Soc. Series C: Appl. Stat.*, 54:207–222, 2005.

G. Feld, D. Randell, and P. Jonathan. On the estimation and application of directional design criteria. *Proc. 38th Int. Conf. on Ocean, Offshore & Arctic Engineering, Scotland*, 2019a.

G. Feld, D. Randell, E. Ross, and P. Jonathan. Design conditions for waves and water levels using extreme value analysis with covariates. *Ocean Eng.*, 173:851–866, 2019b.

P. Jonathan, K. C. Ewans, and G. Z. Forristall. Statistical estimation of extreme ocean environments: The requirement for modelling directionality and other covariate effects. *Ocean Eng.*, 35:1211–1225, 2008.

M. J. Jones, D. Randell, K. Ewans, and P. Jonathan. Statistics of extreme ocean environments: non-stationary inference for directionality and other covariate effects. *Ocean Eng.*, 119:30–46, 2016.

B. D. Marx and P. H. C. Eilers. Direct generalised additive modelling with penalised likelihood. *Comput. Statist. Data Anal.*, 28:193–209, 1998.

D. Randell, K. Turnbull, K. Ewans, and P. Jonathan. Bayesian inference for non-stationary marginal extremes. *Environmetrics*, 27:439–450, 2016.

E. Ross, D. Randell, K. Ewans, G. Feld, and P. Jonathan. Efficient estimation of return value distributions from non-stationary marginal extreme value models using Bayesian inference. *Ocean Eng.*, 142:315–328, 2017.

E. Zanini, E. Eastoe, M. J. Jones, D. Randell, and P. Jonathan. Flexible covariate representations for extremes. *Environmetrics*, 31:e2624, 2020.